Vortex Research Checkpoint

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Status: stopped early at the author's request. The authorized allowance was 30 minutes. Computation was stopped before that limit; resources were not exhausted. Completed results are preserved, while an interrupted run is explicitly excluded.

1. Completed earlier survival study

Eight axisymmetric physical-time runs reached dimensionless time 24, retaining a detected original-field core at every saved sample. They included an unperturbed control, amplitude/phase disturbances, a half-time-step run and saved coarse/larger-domain reference profiles. The medium run's final state differed by 7.37e-8 between time steps 0.025 and 0.0125.

Four selected linear angular disturbances, m=1 through 4, had maximum combined field/velocity norm gains of approximately 1.71, 1.44, 1.43 and 1.43 over time 24. This is finite-time response to selected initial directions, not the complete perturbation spectrum. It neither proves nor disproves full nonlinear stability.

An independent finite-box audit kept relative rotating/moving-frame energy drift below 2.48e-10. Original-field excess-charge changes were accounted for by the integrated discrete boundary balance to absolute residual below 1.87e-9. That energy is not an isolated particle rest mass.

2. New nonlinear angular prototype

A separate research implementation evolves the existing two-field equations on an r/theta/Z grid. It combines radial/axial finite differences with a periodic Fourier angular Laplacian. The main console's physical equations remain unchanged. Four consistency tests passed: axisymmetric reduction, a harmonic angular-Laplacian test, agreement with the prior tangent equations, and differential energy/carrier-charge identities.

Three runs completed: an 8-angle unperturbed control to time 8; an 8-angle m=2 disturbed pilot to time 8; and a 16-angle m=2 disturbed run to time 12. Every sampled angular slice retained a detected core in these runs. The 16-angle case's maximum asymmetry gain was 1.2084, relative energy drift 3.23e-11 and carrier-charge drift 2.77e-11.

The 24-angle run was interrupted without a saved in-progress state. Its output is not a result. Half-step, stronger-disturbance, larger-domain and long-linear tasks in this new batch did not run.

Interpretation and Resume Plan

3. What is established

The preserved moving ring survives the selected finite-time numerical tests, and the new nonlinear angular prototype passes meaningful consistency checks. The saved data provide reproducible evidence for these bounded statements. They do not identify observed matter, establish an isolated mass, demonstrate arbitrary 3D stability, or validate the complete SEFI/DEFI construction.

4. Important unresolved limits

The new nonlinear angular runs do not yet have matched angular-resolution or temporal-refinement comparisons. Their different final times prevent the 8-angle pilots and 16-angle run from serving as a convergence study. Nonlinear Fourier collocation may alias. The m=2 initial disturbance retains discrete symmetry; arbitrary perturbations were not tested. Fixed boundaries and domain dependence remain, and retained grid-cell phase circulation is not a continuum topology theorem.

5. Authorship is not being evolved

The vortex state contains Phi, Sigma and their velocities. It has no independent Authorship coordinate. The original SEFI layer-functional definition, DEFI's evolving internal coordinate and the older software's derived Authorship diagnostics are distinct constructions. A source-grounded variational bridge to the vortex variables must be specified before adding reciprocal Authorship dynamics. Any such experimental extension will be isolated; it will not silently replace the main console equations.

6. Where to resume

First complete matched 16/24-angle and time-step comparisons at the same final time. Then test mixed angular disturbances, spatial/domain sensitivity, radiation and charge budgets, followed by longer physical-time evolution if the numerical checks hold. Separately clarify Authorship's carrier object, coupling functions, dimensions, observables and backreaction. Matter interpretation requires further theoretical and experimental evidence.

7. Preserved records

The repository contains the prototype source, four tests, three completed nonlinear case JSON/NPZ records, the dated batch plan, STOP_RECORD.json and a read-only console evidence snapshot. The earlier survival folder retains eight axisymmetric runs, seven linear angular cases, two balance audits and its verification record. A browser check of the evidence view found eight axisymmetric rows and no JavaScript errors.

Reproduction and provenance

Repository: github.com/JasonSEFIDEFI/SEFI-PY
Folders: research/2026-10-19/vortex-survival and vortex-extended.
Protected physical files: physics.py, ring_operator.py and rest_search.py, verified unchanged against the preceding committed checkpoint.
This report records a paused research checkpoint, not a completed matter model.